FROBENIUS CONSTANTS, STOKES RATIOS, AND THE MIDDLE-ROOT OBSTRUCTION FOR CELLULAR APPROXIMATIONS TO $\zeta(5)$

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Abstract. We compute the Frobenius constants, in the sense of Bloch–Vlasenko, of the order-nine Picard–Fuchs operator attached to the Brown–Zudilin family of cellular approximations to $\zeta(5)$, and we use them to describe an obstruction intrinsic to that family. Three things are new. First, an instrument: for an Apéry-like operator the Frobenius deformation series $\kappa(\varepsilon)$ equals the ratio $S(\varepsilon)/S(0)$ of Stokes constants of the dominant Birkhoff normal solution, the connection point and the growth rate cancelling identically. This replaces a limit converging like $1/n$ by one converging beyond all orders, and delivers 434 digits of κ in seconds. Reproducing the published Golyshev–Zagier and Bloch–Vlasenko tables to 219 digits, we settle two misprints in the latter, stating both the printed and the corrected values. Second, the κ-vector of the Brown–Zudilin operator at its nearest conifold: the identifications $\kappa_2 = -4\zeta(2)$, $\kappa_3 = \frac{550}{87}\zeta(3)$, $\kappa_4 = \frac{365}{29}\zeta(4)$, $\kappa_5 = \frac{514}{87}\zeta(5) - \frac{2200}{87}\zeta(2)\zeta(3)$ — read off by integer-relation detection and verified to 434 digits, not proved — make the primitive part of the first higher Frobenius constant the pure value $\lambda_5 = \frac{514}{87}\zeta(5)$, with the $\zeta(2)\zeta(3)$ impurity exactly the decomposable term $\lambda_2\lambda_3$; the passage from the κ_j to the λ_j is proved, the closed forms themselves are numerical. Third, an identification of the family's purity defect as the Tate class $c = -3/\pi^2 = 12/(2\pi i)^2$, certified to 441 digits, together with closed forms for the connection constants ($A_Q A_{I'} A_I = -\pi^{5/2}/(12\sqrt{37})$, each factor a square root in an S_3 cubic field times a half-integer power of π, with $\Gamma(r/s)$ excluded for every s searched, namely $s \in \{3, 4, 5, 6, 8, 12, 24\}$). We formulate a falsifiable conservation law $r = 1 + \lfloor m/2 \rfloor$ relating the recurrence order to the multiplicity of the exponent 0, verified on four families including a prediction at weight 7; and we apply the resulting picture to show that the weight-7 cellular family admits no pair-worthiness — under the one standing hypothesis $s, \widehat{P} \in \mathrm{Sol}(L)$, which we state but do not prove — its primitive linear form riding the middle characteristic root by a margin of 4.635 nats.

Contents

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1. Introduction

1.1. **Frobenius constants.** Let L be a linear differential operator with regular singular point at the origin, and let $\Phi(s,z) = \sum_{n\geq 0} a_n(s) z^{n+s}$ be its Frobenius deformation, normalised by $a_0 = 1$. Golyshev and Zagier [12] discovered, in the course of proving the gamma conjecture for Fano threefolds of Picard rank one, that the Taylor coefficients of a suitably renormalised limit of Φ are zeta values: for the Apéry operator attached to $\zeta(3)$ they found

$$\kappa_0 = 1, \qquad \kappa_1 = 0, \qquad \kappa_2 = -2\zeta(2), \qquad \kappa_3 = \tfrac{17}{6}\zeta(3). \tag{1}$$

Bloch and Vlasenko [3] gave the invariant definition. If $c \neq 0, \infty$ is a *special reflection point* of L — a regular singular point at which $\sigma_c - 1$ has rank-one image on solutions, and at which all σ_c-invariant solutions of $L^\vee$ are analytic — and γ is a path from 0 to c through regular points, then the *Frobenius constants* $\kappa_{\rho,n}$ are defined by continuing the Frobenius functions $\varphi_{\rho,n} = \frac{1}{n!}\partial_s^n \Phi|_{s=\rho}$ along γ and setting

$$(\sigma_c - 1)\,\varphi_{\rho,n}(t) \;=\; \kappa_{\rho,n}\,\delta(t), \tag{2}$$

where δ spans the one-dimensional image of $\sigma_c - 1$ **[QUOTED]** [3, Def. 22]. The normalisation is $\kappa_0 = 1$; there is no $2\pi i$ and no Γ-factor in it. Bloch and Vlasenko prove that these constants are the expansion coefficients of a *gamma function* attached to $L^\vee$, and that for L of Picard–Fuchs type they lie in the algebra of periods with $2\pi i$ inverted **[QUOTED]** [3, Thm. 30, Cor. 31]. Kerr [14] recasts the construction in terms of unipotent extensions; Roy and Vlasenko [22] compute it for families of elliptic curves, where the higher κ's become iterated integrals of modular forms; Golyshev, Kerr and Sasaki [11] relate the adjacent notion of Apéry limits to extension classes. The relation with Golyshev's *Apéry constants* [10] is one of family resemblance only: those are limits $\lim b_n/a_n$, a different invariant, and we shall keep the two apart.

A structural feature of (2) governs everything below. If $m = m(\rho)$ is the multiplicity of ρ as a root of the indicial polynomial, then $\varphi_{\rho,0}, \ldots, \varphi_{\rho,m-1}$ are honest solutions of L, while $\varphi_{\rho,k}$ for $k \geq m$ solves an inhomogeneous equation. Bloch and Vlasenko put it as follows **[QUOTED]** [3, p. 21]:

> "…the Frobenius constants corresponding to actual solutions of L (with $k < m(\rho)$) are periods of the limiting Hodge structure at $t = 0$. However, there is no reason to expect that the operators $(D-\rho)^j L$ with $j > 0$ are geometric. From this point of view, it is surprising that the higher Frobenius constants in the above examples are periods."

The constants $\kappa_0, \ldots, \kappa_{m-1}$ are therefore *geometric*, and κ_m is the first *higher* one. For the Apéry $\zeta(3)$ operator $m = 3$ and the first higher constant is $\kappa_3 = \frac{17}{6}\zeta(3)$ — a pure rational multiple of $\zeta(3)$, of weight exactly 3.

1.2. **What was not known beyond rank two.** All the operators for which Frobenius constants have been tabulated in the literature are of order 2, 3 or 4 with a maximally unipotent monodromy point at the origin: the Apéry operators, Beauville's six stable elliptic families [22], the 17 rank-one Fano families [12], and the hypergeometric Calabi–Yau operators, for which κ

has a closed form in Γ-quotients **[QUOTED]** [3, Prop. 26]. In every tabulated case the multiplicity of the exponent 0 is at most 4, so the first higher constant has weight at most 4, and the values $\zeta(5), \zeta(7), \dots$ appear only in the tail, as *higher* constants of a low-weight operator: for instance $\kappa_5 = \frac{7}{3}\zeta(5) - \frac{17}{3}\zeta(2)\zeta(3)$ for the Apéry $\zeta(3)$ operator, and $\kappa_5 = \zeta(5) - 3\zeta(2)\zeta(3)$ for the Apéry $\zeta(2)$ one. In both, $\zeta(5)$ arrives mixed with $\zeta(2)\zeta(3)$.

Two questions were consequently open.

(a) *Is there an operator whose* first *higher Frobenius constant is a pure rational multiple of* $\zeta(5)$*?* A search of the literature turned up none; the slot appeared to be empty.
(b) *Do the Frobenius constants say anything about irrationality constructions?* The κ's are attached to a differential operator, whereas irrationality proofs are driven by the arithmetic of a recurrence. The two are the same datum viewed twice; but the dictionary had not been used in this direction.

The second question is worth stating against the background of what is known. Apéry's proof that $\zeta(3)$ is irrational [1, 2], presented in [7], remains the only such result for an odd zeta value; nothing is known about any single $\zeta(2k+1)$ with $k \geq 2$. Infinitely many odd zeta values are irrational [20, 21], at least $2^{(1-\epsilon)\log s/\log\log s}$ among $\zeta(3), \dots, \zeta(s)$ [9]; at least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational [24, 25], at least two of $\zeta(5), \dots, \zeta(35)$ [16], and one of $\zeta(5), \dots, \zeta(25)$ by elementary means [26]. Structural reformulations have come from Siegel's lemma [8], from transfinite diameter [5], and from the motivic viewpoint [4]. What all the constructive proofs share is a recurrence and a saddle; the results below concern which saddle a *pure* linear form is allowed to occupy.

The natural test object for both questions is the Brown–Zudilin family. Brown [4] interpreted Apéry-type irrationality proofs as period computations on moduli spaces $\mathcal{M}_{0,n}$; Brown and Zudilin [6] constructed from cellular integrals on $\mathcal{M}_{0,8}$ an effective sequence of rational approximations to $\zeta(5)$ with $|\zeta(5) - p/q| < q^{-0.86}$, the best effective exponent known. Their integrals decompose as

$$\begin{gathered} I \;=\; Q \cdot \bigl(2\zeta(5) + 4\zeta(3)\zeta(2)\bigr) - 4\widehat{P} \cdot \zeta(2) - 2P \;=\; 2I' + 4\,I''\zeta(2), \\ I' = Q\zeta(5) - P, \qquad I'' = Q\zeta(3) - \widehat{P}, \end{gathered} \tag{3}$$

and $Q, P, \widehat{P}$ satisfy a third-order Apéry-like recurrence with characteristic polynomial $4\lambda^3 - 2368\lambda^2 - 188\lambda + 1$ **[QUOTED]** [6, §2]. The whole difficulty of the family is visible in (3): the *pure* linear forms I' and I'' ride the middle root λ_2, while the smallest root λ_1 — the one that would give worthiness exceeding 1, hence irrationality — is monopolised by the *impure* combination I, whose leading period $\zeta(5) + 2\zeta(2)\zeta(3)$ has depth 2. We call this the *middle-root obstruction.* The purpose of this paper is to compute the Frobenius data of the operator behind (3), to identify the constant that measures the obstruction, and to test whether the obstruction is a feature of the particular family or of its weight.

1.3. Results.

(1) The instrument (§2). We prove (Theorem 2.3) that for an Apéry-like operator with a strictly dominant simple characteristic root λ, and with $c = 1/\lambda$ the nearest singularity,

$$\kappa(\varepsilon) \;=\; \frac{S(\varepsilon)}{S(0)}, \qquad S(\varepsilon) := \lim_{n\to\infty} \frac{a_n(\rho+\varepsilon)}{A(n+\varepsilon)},$$

where A is the dominant Birkhoff normal solution. The factors c^{ε} and λ^{ε} of [3, Lemma 24] cancel identically. The resulting algorithm converges beyond all orders and gives 434 digits where the direct implementation of [3, Lemma 24] gives nine.

(2) Validation and two errata (§3). We reproduce the published κ- and λ-tables of [12, 13, 3, 22] at 219 digits and record two misprints in [3], quoting the printed and the corrected value in each case, and give new values $\kappa_8, \kappa_9, \lambda_8, \lambda_9$ for the Apéry $\zeta(2)$ operator.

(3) The κ-vector of the Brown–Zudilin operator, and a pure λ_5 (§§4–5). The operator has order 9, with the exponent 0 occurring with multiplicity $m = 5$ at the origin, and all three finite singularities are conifolds with exponent set $\{0, \ldots, 7\} \cup \{3/2\}$. At the nearest conifold, **[VERIFIED:** 434 **digits]**

$$\kappa_2 = -4\zeta(2), \quad \kappa_3 = \tfrac{550}{87}\zeta(3), \quad \kappa_4 = \tfrac{365}{29}\zeta(4), \quad \kappa_5 = \tfrac{514}{87}\zeta(5) - \tfrac{2200}{87}\zeta(2)\zeta(3),$$

whence, **[PROVED]** given those identifications, $\lambda_5 = \frac{514}{87}\zeta(5)$, filling the slot (a). The impurity of κ_5 is precisely the decomposable term $\lambda_2\lambda_3$.

(4) The purity defect is Tate (§7). The constant c that rotates the middle-ray pair (I', I'') onto the minimal ray is $c = -1/(2\zeta(2)) = -3/\pi^2 = 12/(2\pi i)^2$, proved from (3) and certified against exact ladders to 441 digits.

(5) A conservation law (§8). We conjecture that for Apéry-like families the order r of the recurrence and the multiplicity m of the exponent 0 satisfy $r = 1 + \lfloor m/2 \rfloor$, the rays being indexed by the depth of the period they carry, and we verify it on four families, the fourth (weight 7, $m = 7$, $r = 4$) being a prediction made before the computation of m.

(6) Connection constants are $\Gamma(1/2)$-monomials (§9). We identify all four Brown–Zudilin connection constants: $A_Q A_{I'} A_I = -\pi^{5/2}/(12\sqrt{37})$, with each factor a square root in the S_3 cubic field of the singular locus times a half-integer power of π, and with $\Gamma(r/s)$ excluded for every denominator searched, $s \in \{3, 4, 5, 6, 8, 12, 24\}$.

(7) No pair-worthiness at weight 7 (§10). For the weight-7 cellular family the primitive form I' satisfies $I' = I + \zeta(2)|I''|$ termwise, a sum of two positive quantities; hence — given Theorem 10.1's standing hypothesis $s, \widehat{P} \in \mathrm{Sol}(L)$, which §10.7 does not prove — it cannot decay faster than I'', and the fast sector is excluded without any cancellation argument. The measured denominator budget is $\kappa = 7$, tight, so $\gamma_{5,7} \leq 0.3379$ and the shortfall is 4.635 nats, a factor 103 per step.

1.4. **Evidence classes and conventions.** Because a substantial part of what follows is high-precision computation, every assertion in this paper carries exactly one of the following markers.

- **[PROVED]** — a complete derivation is given, or the statement is an exact symbolic identity verified in exact arithmetic (integers, rationals, or polynomial algebra).
- **[VERIFIED:** d **digits]** — a high-precision computation agreeing with the stated closed form to d decimal digits. For an integer-relation identification we state in addition the detection tolerance and the coefficient bound within which the relation was searched.
- **[EXCLUDED: tol, bound]** — an integer-relation search returned no relation at the stated tolerance and coefficient bound. An exclusion is a statement about a finite search box and nothing more.
- **[QUOTED]** — a statement copied from a source, with the location; we have re-read the source at that location.
- **[OPEN]** — not settled here.

Numerical agreement is never presented as proof. Where a quantity is quoted to d digits, d is the observed self-agreement between two independent runs (differing in the number of terms n, in the truncation order, or in the working precision), not the working precision itself.

We write $\theta = z\,\mathrm{d}/\mathrm{d}z$, and an Apéry-like operator in the form $L = \sum_{j=0}^{d} z^j q_j(\theta)$ with $q_j \in \mathbb{Q}[x]$; the indicial polynomial at $z = 0$ is $I(s) = q_0(s)$, and the Frobenius recursion reads

$\sum_{j=0}^{n} q_j(n+s-j)\, a_{n-j}(s) = 0$ with $a_0 = 1$. We write $\zeta_2 = \zeta(2)$ where space demands. Following [6] we write I'' for the $\zeta(3)$-form; the notation $\widehat{I}$ is used interchangeably.

Remark 1.1 (Methodology; flagged). The computations reported here were carried out with substantial machine assistance, including automated search over candidate closed forms. Every identification quoted below was re-derived and re-checked independently of the search that proposed it, and the precision, tolerance and coefficient bound of each check are stated at the point of use; the exclusions in §6 and §9 record searches that returned nothing, and are reported with the same care as the positive findings. Appendix B lists the computational record.

2. The instrument: κ as a ratio of Stokes constants

2.1. **The computational form of the definition.** We recall the recipe of [3, Lemma 24], in the form in which we shall use it **[QUOTED]** [3, Lemma 24]: suppose the special reflection point $t = c$ is the singularity nearest 0, that $c \in \mathbb{R}_{>0}$, that $|\varphi_{\rho,0}(t)| \to \infty$ as $t \to c^-$, that $a_n(\rho)$ is eventually of one sign, and that $\lambda_k := \lim_n a_n^{(k)}(\rho)/(k!\, a_n(\rho))$ exists. Then $\kappa_{\rho,0} \neq 0$ and

$$\frac{\kappa_{\rho,k}}{\kappa_{\rho,0}} = \sum_{j=0}^{k} \frac{(\log c)^j}{j!}\, \lambda_{k-j}. \tag{4}$$

Equivalently, with $\Lambda(\varepsilon) := \lim_{n\to\infty} a_n(\rho+\varepsilon)/a_n(\rho)$,

$$\kappa(\varepsilon) = \sum_{k\geq 0} \kappa_{\rho,k}\varepsilon^k = c^{\varepsilon}\, \Lambda(\varepsilon). \tag{5}$$

This is also, verbatim, the recipe of [12, §9], there written $\kappa_0(\varepsilon) = C^{-\varepsilon} \lim_n A_n(\varepsilon)/A_n$ with $C = 1/c$ the exponential growth rate.

Taken literally, (5) is a poor algorithm: $a_n(\rho+\varepsilon)/a_n(\rho)$ converges like $1/n$, and Richardson extrapolation on it reaches about nine digits before the conditioning of the extrapolation defeats the gain. Golyshev and Zagier already observed **[QUOTED]** [12, §9] that dividing instead by the Birkhoff–Trjitzinsky asymptotic $A(n+\varepsilon)$, rather than by $a_n(\rho)$, converges faster than any power of n, and reported 300 digits from $n = 100$ in the Apéry case. Theorem 2.3 explains why, and shows that the correction is not a numerical trick but an identity: the two exponential factors in (5) cancel exactly, leaving κ as a pure ratio of Stokes constants.

2.2. **The Birkhoff normal solution.** Fix an operator $L = \sum_j z^j q_j(\theta)$ of order r and degree d in z, and put $D = \max_j \deg q_j$. The characteristic polynomial of the associated recurrence is

$$\chi(\lambda) = \lambda^d \sum_j \lambda^{-j}\, [x^D] q_j(x),$$

whose roots are the reciprocals of the finite singularities of L. Assume one root λ is simple and strictly dominant in absolute value.

Proposition 2.1 (Birkhoff normal solution). **[PROVED]** *There is a unique formal expression*

$$A(x) = \lambda^x\, x^\alpha\, F(1/x), \qquad F(u) = \sum_{k\geq 0} c_k u^k, \quad c_0 = 1, \tag{6}$$

satisfying $\sum_j q_j(x-j)A(x-j) = 0$ *as a formal series in* $u = 1/x$. *Writing* $Q_j(u) := u^D q_j(1/u - j) \in \mathbb{Q}[u]$ *and*

$$\beta := \sum_j \lambda^{-j} Q_j'(0), \qquad \gamma := \sum_j \lambda^{-j}\, j\, Q_j(0), \qquad H_k(u) := \sum_j \lambda^{-j} Q_j(u)(1-ju)^{\alpha-k},$$

one has $\sum_j \lambda^{-j} Q_j(0) = 0$ *(the characteristic equation),*

$$\alpha = \beta/\gamma, \qquad \text{and} \qquad c_m = -\frac{1}{\gamma m} \sum_{k<m} c_k\, h_{k,\, m+1-k}, \qquad h_{k,i} := [u^i] H_k(u). \tag{7}$$

Proof. Substituting (6) into $\sum_j q_j(x-j)A(x-j) = 0$, dividing by $\lambda^x x^\alpha$ and multiplying by u^D with $u = 1/x$ turns the equation into $\sum_j \lambda^{-j} Q_j(u)(1-ju)^\alpha F\big(u/(1-ju)\big) = 0$, since $q_j(x-j) = u^{-D} Q_j(u)$ and $1/(x-j) = u/(1-ju)$. Expanding $F(u/(1-ju)) = \sum_k c_k u^k (1-ju)^{-k}$ gives $\sum_{k\ge 0} c_k\, u^k H_k(u) = 0$. The coefficient of u^0 is $c_0 H_0(0)$, and $H_k(0) = \sum_j \lambda^{-j} Q_j(0)$ is independent of k; so this coefficient vanishes precisely when λ solves the characteristic equation, and then $h_{k,0} = 0$ for all k. The coefficient of u^1 is therefore $c_0 h_{0,1} = \beta - \alpha\gamma$, which forces $\alpha = \beta/\gamma$. For $m \ge 1$ the coefficient of u^{m+1} is $\sum_{k\le m} c_k h_{k,m+1-k}$, whose top term is $c_m h_{m,1} = c_m(\beta - (\alpha - m)\gamma) = c_m\, m\gamma$; solving for c_m gives (7). Simplicity of λ gives $\gamma \ne 0$, so the recursion determines all c_m uniquely. □

Remark 2.2. The exponent α comes out an *exact rational* and is a strong self-check on the implementation: $\alpha = -1$ for the Apéry $\zeta(2)$ operator, $\alpha = -3/2$ for the Apéry $\zeta(3)$ operator, $\alpha = -5/2$ for the Brown–Zudilin operator. In each case $\alpha = -\rho - 1$ with ρ the non-integer local exponent at the conifold, i.e. $\rho = 0,\ 1/2,\ 3/2$ respectively. **[PROVED]** (exact rational arithmetic; for the Brown–Zudilin case the value $\rho = 3/2$ is proved independently in Proposition 4.2).

2.3. The Stokes-ratio identity.

Theorem 2.3 (Stokes-ratio form of κ). **[PROVED]** *Let L be as above, with λ a simple strictly dominant characteristic root and A the Birkhoff normal solution of Proposition 2.1. Let ρ be a local exponent at $z = 0$ for which the limit*

$$S(\varepsilon) \ := \ \lim_{n\to\infty} \frac{a_n(\rho+\varepsilon)}{A(n+\varepsilon)}$$

exists and is non-zero for ε in a neighbourhood of 0. *Let $c = 1/\lambda$ be the nearest singularity, assumed a special reflection point, and suppose $\kappa(\varepsilon) = c^\varepsilon \Lambda(\varepsilon)$ as in* (5). *Then*

$$\boxed{\kappa(\varepsilon) \ = \ \frac{S(\varepsilon)}{S(0)}} \tag{8}$$

identically as formal power series in ε.

Proof. The Frobenius deformation acts on the recursion by the substitution $n \mapsto n+s$: the sequence $\big(a_n(\rho+\varepsilon)\big)_n$ satisfies $\sum_j q_j\big((n+\varepsilon)-j\big) a_{n-j}(\rho+\varepsilon) = 0$ when $\rho = 0$, and in general the same equation with $n+\rho+\varepsilon$ in place of $n+\varepsilon$. Hence the *same* formal solution A of Proposition 2.1, evaluated at the shifted argument $x = n+\rho+\varepsilon$, governs its asymptotics; absorbing ρ into the normalisation we write $A(n+\varepsilon)$. Therefore

$$\Lambda(\varepsilon) = \lim_{n\to\infty} \frac{a_n(\rho+\varepsilon)}{a_n(\rho)} = \frac{S(\varepsilon)}{S(0)} \cdot \lim_{n\to\infty} \frac{A(n+\varepsilon)}{A(n)}.$$

By (6), $A(n+\varepsilon)/A(n) = \lambda^\varepsilon\,(1+\varepsilon/n)^\alpha\, F(1/(n+\varepsilon))/F(1/n) \to \lambda^\varepsilon$. Thus $\Lambda(\varepsilon) = \lambda^\varepsilon S(\varepsilon)/S(0)$ and $\kappa(\varepsilon) = c^\varepsilon \Lambda(\varepsilon) = \lambda^{-\varepsilon}\lambda^\varepsilon S(\varepsilon)/S(0) = S(\varepsilon)/S(0)$. □

Remark 2.4 (Why this is fast). The error in $a_n(\varepsilon)/A(n+\varepsilon) - S(\varepsilon)$ has two parts: the contribution of the subdominant solutions, of size $O\big((\lambda_{\text{next}}/\lambda)^n\big)$, and the truncation of the divergent series F after M terms, of size $\approx M!/(\mathcal{S}n)^M$ with $\mathcal{S} = \log|\lambda/\lambda_{\text{next}}|$. Both are governed by the

Table 1. The truncation model of Remark 2.4 against observed self-agreement. $\mathcal{S} = \log|\lambda/\lambda_{\mathrm{next}}|$; predicted $\log_{10}$ error is $\log_{10}\big(M!/(\mathcal{S}n)^M\big)$. **[VERIFIED: 208–434 digits, as tabulated]**

operator	$\mathcal{S}$	n	M	predicted digits	observed
Apéry $\zeta(3)$	7.0510	300	130	212	208–210
Apéry $\zeta(3)$	7.0510	400	130	229	(capped by dps = 220)
Brown–Zudilin	8.8560	700	260	469	434
Brown–Zudilin	8.8560	800	260	485	(capped by dps = 520)

same $\mathcal{S}$, and optimal truncation $M \approx \mathcal{S}n$ gives error $e^{-\mathcal{S}n}$: the convergence is beyond all orders. For the Apéry $\zeta(3)$ operator $\mathcal{S} = 2\log(17+12\sqrt{2}) = 7.05099\ldots$, so $n = 100$ yields $7.05099 \cdot 100/\log 10 = 306.2$ digits — reproducing exactly the "300 digits from $n = 100$" of [12, §9]. **[PROVED]** (as an asymptotic count; the observed precisions in §3 and §5 agree with the model, see Table 1.)

2.4. **Relation to the p-adic side, stated without overreach.** There is a p-adic story running parallel to this one, and it is easy to overstate the parallel. On the p-adic side, the limits of Frobenius towers attached to the same families are *values* of $G(x) = \Gamma_p(x)e^{-\Gamma_p'(0)x} = \exp\big(-\sum_{m\ge 2}\zeta_p(m)x^m/m\big)$ at p-power arguments, whereas the Frobenius *structure* constants of [27] are the Taylor coefficients of the same G at 0. What is verified here, and all that we assert, is:

(a) the archimedean series $\log\kappa(\varepsilon) = \sum_j \lambda_j\varepsilon^j$ has $\lambda_j \in \mathbb{Q}\cdot\zeta(j)$ for $j \le 5$ on the Brown–Zudilin and Apéry $\zeta(3)$ operators, and for $j \le 4$ on the Apéry $\zeta(2)$ operator — an initial segment of exactly the shape $\exp\big(-\sum_m r_m\zeta(m)x^m\big)$, $r_m \in \mathbb{Q}$;
(b) the archimedean connection constants are $\Gamma(1/2)$-monomials times algebraic numbers (§9).

We did *not* find, and do not claim, a literal identity between the two Γ's. The honest statement is that both sides are expansion data of a Γ-type generating function attached to the operator — Γ_p on one side, the Γ_{ξ_0} of **[QUOTED]** [3, Thm. 30] on the other — and that in both cases the coefficients are (multiple) zeta values.

3. Validation, and two errata

Reproducing the published values is the credibility of everything that follows, and we report it in full.

3.1. **Controls.** Run parameters: working precision dps = 220, ε-truncation $K = 13$, Birkhoff truncation $M = 130$, $n \in \{300, 400\}$; self-agreement across n is 208–210 digits, in agreement with the model of Remark 2.4 (Table 1).

Remark 3.1 (The two published packagings agree). **[VERIFIED: 60 digits]** The tables of [12] and [13] present the same data twice: [13, (47)] lists κ_j for $j \le 11$, [12, §9] lists $\lambda_j = [\varepsilon^j]\log\kappa(\varepsilon)$ for the same range. Exponentiating the λ-list reproduces the κ-list, and vice versa, to 60 digits — the limit here being the number of digits carried by the printed rational coefficients, not the computation. Two re-packagings we record because they are convenient and are not printed in that form: $\lambda_6 = -\frac{2}{3}\zeta(6) - \frac{1}{72}\zeta(3)^2 = -\frac{16}{105}\zeta(2)^3 - \frac{1}{72}\zeta(3)^2$ and $\lambda_8 = \frac{29}{12}\zeta(8) - \frac{11}{18}\zeta(3)\zeta(5) = \frac{58}{175}\zeta(2)^4 - \frac{11}{18}\zeta(3)\zeta(5)$. **[PROVED]** (exact rational identities $\zeta(6) = \frac{8}{35}\zeta(2)^3$, $\zeta(8) = \frac{24}{175}\zeta(2)^4$.)

TABLE 2. Instrument validation. All residuals are the difference between the computed value and the published closed form; each source location was re-read **[QUOTED]**. **[VERIFIED: 219 digits]**

control	source	result
Apéry $\zeta(3)$: exponent α	—	$-3/2$, exact rational
Apéry $\zeta(3)$: Stokes constant $S(0)$	classical	$0.22004376711264303785068975 98\ldots$, equal to $(1+\sqrt{2})^2/(2^{9/4}\pi^{3/2})$ to all digits computed
Apéry $\zeta(3)$: $\kappa_2, \kappa_3, \kappa_4$	[3, Ex. 29]	residuals $< 10^{-219}$ against $-2\zeta(2)$, $\frac{17}{6}\zeta(3)$, $2\zeta(4)$; for κ_5 see Erratum 3.2
Apéry $\zeta(3)$: $\kappa_6, \ldots, \kappa_{10}$	[13, (47)]	residuals $< 10^{-219}$
Apéry $\zeta(3)$: $\lambda_2, \ldots, \lambda_{10}$	[12, §9]	all reproduced; the two published packagings are mutually consistent, see Remark 3.1
Apéry $\zeta(2)$: $\kappa_2, \ldots, \kappa_7$	[3, Ex. 28], [22, Tab. 2 D]	residuals $< 10^{-219}$, with the one exception recorded in Erratum 3.3

3.2. **Two errata in** [3]**.** Both printed forms below were read verbatim from the published text before the computation was run; both corrections agree with [13, (47)] and [22, Table 2] respectively. We state them plainly and with no implication beyond a typesetting slip.

Observation 3.2 (Erratum 1)**. [VERIFIED: 219 digits]** [3, Example 29] prints, for the Apéry $\zeta(3)$ operator,

$$\kappa_5 = \tfrac{7}{5}\zeta(5) - \tfrac{17}{3}\zeta(2)\zeta(3) \qquad \textbf{[QUOTED]}\text{ [3, Ex. 29].}$$

The computed value is

$$\kappa_5 = -8.785226556350148175010298033349298516332336165569 63\ldots$$

The form $\frac{7}{3}\zeta(5) - \frac{17}{3}\zeta(2)\zeta(3)$ agrees with it to 4.6×10^{-219}; the printed form differs by 0.9678. The correct value is therefore

$$\kappa_5 = \tfrac{7}{3}\zeta(5) - \tfrac{17}{3}\zeta(2)\zeta(3),$$

in agreement with $\kappa_5 = -\frac{17}{18}\pi^2\zeta(3) + \frac{7}{3}\zeta(5)$ of **[QUOTED]** [13, (47)]. The erratum concerns the coefficient of $\zeta(5)$ only; the term $-\frac{17}{3}\zeta(2)\zeta(3)$ stands as printed. All other values listed in [3, Ex. 29] check out.

Observation 3.3 (Erratum 2)**. [VERIFIED: 219 digits]** [3, Example 28] prints, for the Apéry $\zeta(2)$ operator,

$$\kappa_6 = \tfrac{87}{16}\zeta(6) - \tfrac{5}{2}\zeta(3)^2 \qquad \textbf{[QUOTED]}\text{ [3, Ex. 28].}$$

The computed value is

$$\kappa_6 = 9.14415489562452778198190394025559993842547\ldots$$

The form with $+\frac{5}{2}\zeta(3)^2$ agrees to 5.3×10^{-220}; the printed form differs by 7.2247. The correct value is

$$\kappa_6 = \tfrac{87}{16}\zeta(6) + \tfrac{5}{2}\zeta(3)^2,$$

in agreement with **[QUOTED]** [22, Table 2, case D]. The remaining entries $\kappa_2, \kappa_3, \kappa_4, \kappa_5, \kappa_7$ of [3, Ex. 28] check out.

3.3. **New values for the Apéry $\zeta(2)$ operator.** The published tables for this operator stop at κ_7. We add

$$\begin{aligned}
\kappa_8 &= \tfrac{8627}{4200}\zeta(2)^4 - \tfrac{7}{2}\zeta(2)\zeta(3)^2 + \tfrac{15}{2}\zeta(3)\zeta(5),\\
\kappa_9 &= \tfrac{29}{21}\zeta(2)^3\zeta(3) + \tfrac{5}{3}\zeta(3)^3 - \tfrac{19}{4}\zeta(2)^2\zeta(5) + \tfrac{115}{16}\zeta(2)\zeta(7) - \tfrac{2011}{72}\zeta(9),\\
\lambda_8 &= \tfrac{336593}{105000}\zeta(2)^4 + \tfrac{2}{5}\zeta(2)\zeta(3)^2 + \tfrac{11}{2}\zeta(3)\zeta(5),\\
\lambda_9 &= -\tfrac{14263}{5250}\zeta(2)^3\zeta(3) - \tfrac{2}{3}\zeta(3)^3 - \tfrac{649}{100}\zeta(2)^2\zeta(5) - \tfrac{39}{16}\zeta(2)\zeta(7) - \tfrac{2011}{72}\zeta(9).
\end{aligned} \tag{9}$$

[VERIFIED: 219 digits]. As an internal check, the λ-values were re-derived from the κ-values by exponentiation — an exact algebraic consequence of the κ-identifications **[PROVED]** — and agree numerically to the full precision of the printed rational data (60 digits). In the same spirit, $\lambda_4 = -\frac{39}{20}\zeta(4)$ and $\lambda_5 = \zeta(5) - \frac{1}{5}\zeta(2)\zeta(3)$ for this operator, both exact consequences of $\kappa_2 = -\frac{7}{5}\zeta(2)$, $\kappa_3 = 2\zeta(3)$, $\kappa_4 = \frac{1}{2}\zeta(4)$, $\kappa_5 = \zeta(5) - 3\zeta(2)\zeta(3)$ **[PROVED]**. We did not attempt κ_{10}, which requires $\zeta(3,7)$ and $\zeta(2)\zeta(3,5)$ and hence a validated high-precision multiple-zeta-value routine **[OPEN]**.

4. The Brown–Zudilin operator

4.1. **From the recurrence to the operator.** Brown and Zudilin give the order-three recurrence satisfied by I_n, hence by Q_n, P_n, $\widehat{P}_n$, obtained by creative telescoping from the cellular integrals evaluated with [17] and [15] **[QUOTED]** [6, §2]; a third-order Apéry-like recursion for $\zeta(5)$ of this shape goes back to [23]:

$$A(n)u_{n+1} + B(n)u_n + C(n)u_{n-1} + D(n)u_{n-2} = 0, \tag{10}$$

with

$$\begin{aligned}
A(n) &= 2(2n+1)(41218n^3 - 48459n^2 + 20010n - 2871)(n+1)^5,\\
B(n) &= -\big(97604224n^9 + 178061760n^8 + 72005308n^7 - 48634688n^6 - 39076836n^5\\
&\qquad + 2622730n^4 + 7581006n^3 + 920112n^2 - 543402n - 120582\big),\\
C(n) &= -2n\big(3874492n^8 - 2617900n^7 - 3144314n^6 + 2947148n^5 + 647130n^4\\
&\qquad - 1182926n^3 + 115771n^2 + 170716n - 44541\big),\\
D(n) &= n(41218n^3 + 75195n^2 + 46746n + 9898)(n-1)^5.
\end{aligned}$$

With $\theta = z\,\mathrm{d}/\mathrm{d}z$ and $Y(z) = \sum_n u_n z^n$ this transports to

$$L \;=\; A(\theta-1) + z\,B(\theta) + z^2\,C(\theta+1) + z^3\,D(\theta+2), \tag{11}$$

of order 9 in θ and degree 3 in z. **[PROVED]** (exact symbolic computation). The residual of (10) is exactly 0 for $u \in \{Q, P, \widehat{P}\}$ at every $n = 2, \dots, 359$ (1074 exact rational checks), with anchors matching the printed values $Q = 1, 21, 2989$; $P = 0, \frac{87}{4}, \frac{1190161}{384}$; $\widehat{P} = 0, \frac{101}{4}, \frac{344923}{96}$ of **[QUOTED]** [6, §2]. **[VERIFIED: exact rational arithmetic, 1074 checks; no digits are at issue]**. It also holds at $n = 1$ for Q and $\widehat{P}$ but not for P, which carries an inhomogeneity there.

4.2. **Local data.**

Proposition 4.1. [PROVED] *(exact symbolic computation.) For the operator* (11)*:*

(i) the coefficient of θ^9 is $41218\,(z^3 - 188z^2 - 2368z + 4)$, the reciprocal of the characteristic polynomial $4\lambda^3 - 2368\lambda^2 - 188\lambda + 1$; the finite nonzero singularities are $z_i = 1/\lambda_i$;

(ii) the indicial polynomial at $z = 0$ *is*

$$I(s) = q_0(s) = 2\,s^5(2s-1)\bigl(41218s^3 - 172113s^2 + 240582s - 112558\bigr),$$

so the exponent 0 *has multiplicity* $m = 5$*, and the other exponents are* $1/2$ *and the three roots of an irreducible cubic;*

(iii) the z^3 *part is* $(\theta+1)^5(\theta+2)(41218\theta^3 + 322503\theta^2 + 842142\theta + 733914)$*, the same fivefold structure at* $z = \infty$*;*

(iv) $\operatorname{disc}(z^3 - 188z^2 - 2368z + 4) = 2^4 \cdot 37^3 \cdot 557^2$*, not a square, so the cubic field* $\mathbb{Q}(\lambda_3)$ *has Galois group* S_3*; note* $41218 = 2 \cdot 37 \cdot 557$*.*

The point $z = 0$ is thus *not* a MUM point for the order-9 operator — that would require multiplicity 9 — but it carries a rank-5 maximally unipotent sub-block, with local logarithmic depth up to $\log^4 z$. The multiplicity equals the motivic weight, $m = 5$. This resolves an apparent tension in the literature-adjacent folklore: the *recurrence* (10) has order 3, and one might infer a local log-depth of at most 2, too little room for weight 5; but the *operator* has a rank-5 unipotent block, which is exactly enough.

Proposition 4.2 (All three finite singularities are conifolds)**. [PROVED]** *Let* c_9, c_8 *be the coefficients of* θ^9, θ^8 *in* (11)*. At a simple root* z_0 *of* c_9 *the local exponents of an order-*9 *operator are* $\{0, 1, \dots, 7\} \cup \{\rho\}$ *with* $\rho = 8 - c_8(z_0)/\bigl(z_0 c_9'(z_0)\bigr)$*. Here*

$$\rho(z) = \frac{-679z^3 + 106784z^2 + 1082176z - 1384}{74z(-3z^2 + 376z + 2368)},$$

and $\rho(z) - \frac{3}{2}$ *has numerator* $-173\,(z^3 - 188z^2 - 2368z + 4)$*, a nonzero rational multiple of the singular cubic. Hence* $\rho \equiv \frac{3}{2}$ *identically on the singular locus:*

all three singularities z_1, z_2, z_3 *are conifold points with exponents* $\{0, 1, \dots, 7\} \cup \{3/2\}$*.*

This is an exact algebraic identity, not a numerical fit, and it has a structural payoff: exactly one non-integer exponent at each singular point means that $\sigma_c - 1$ has rank-one image there, so the first condition of **[QUOTED]** [3, Def. 21] holds at all three singularities. It also confirms $\alpha = -\rho - 1 = -5/2$ for every ray, matching the measured value of Remark 2.2.

4.3. **Applicability of** [3, Def. 22]**, and one hypothesis that fails.** We record precisely what holds and what does not.

- *Reflection point, first condition:* this holds at all three of the singularities, by Proposition 4.2. **[PROVED]**
- *Reflection point, second condition,* namely analyticity at c of the σ_c-invariant solutions of $L^\vee$: not verified here. **[OPEN]**
- *Nearest singularity real and positive:* one has $z_3 = 1/\lambda_3 = 0.0016889627\ldots$, so the frame of (4) applies. **[PROVED]**
- $a_n(0)$ *of one sign:* $a_n(0) = Q_n = 1, 21, 2989, 714549, \ldots$, verified exactly from the Frobenius recursion — which is simultaneously the check that the q_j are correct. **[VERIFIED: exact, $n \le 359$]**
- *Hypothesis* $|\varphi_{\rho,0}(t)| \to \infty$ *as* $t \to c^-$*: fails.* With $\alpha = -5/2$ one has $\sum_n Q_n z_3^n \asymp \sum n^{-5/2} < \infty$. It *also* fails for the Apéry $\zeta(3)$ operator ($\alpha = -3/2$), for which [3] nonetheless quote the conclusion and for which we verify it to 219 digits in §3.

Observation 4.3. [VERIFIED: 219 digits at $\alpha = -3/2$; consistency checks at $\alpha = -5/2$, see §5] Empirically, hypothesis $|\varphi_{\rho,0}(t)| \to \infty$ of [3, Lemma 24] is *sufficient but not necessary*: the conclusion $\kappa(\varepsilon) = c^\varepsilon \Lambda(\varepsilon)$ holds at $\alpha = -3/2$ and at $\alpha = -5/2$. This is the licence under which we run the machinery on the Brown–Zudilin operator. It is an empirical licence, and we flag it as such.

Finally, the rank-5 block is a feature and not an obstruction. Definition 22 of [3] never requires MUM at $z = 0$; it requires a reflection point. What $m(0) = 5$ does is decide *which* κ_j is the first higher one: $\kappa_0, \dots, \kappa_4$ are periods of the limiting mixed Hodge structure, and κ_5 is the first non-geometric constant — of weight 5. That is exactly why this operator is the right place to look for the $\zeta(5)$-analogue of $\kappa_3 = \frac{17}{6}\zeta(3)$.

5. The κ-vector, and a pure primitive λ_5

5.1. **The computation.** Run parameters: dps = 520, ε-truncation $K = 16$, Birkhoff truncation $M = 260$, $n \in \{700, 800\}$; self-agreement across n is 434 digits, consistent with the truncation model (Table 1) and capped by the working precision. As a cross-check of the entire pipeline, the Stokes constant

$$S(0) = 0.06667642572715676784165334063934544446962874559245 6\dots$$

reproduces the connection constant A_Q of §9, computed by Neville extrapolation in an entirely separate implementation, in every digit of the latter's 158-digit stability window. **[VERIFIED: 158 digits, independent implementations]**

Theorem 5.1 (The Brown–Zudilin κ-vector). **[VERIFIED: 434 digits; integer-relation detection at tolerance 10^{-380} with coefficient bound 10^{14}; residuals 10^{-440} to 10^{-451} in the working session, which the archived 420-digit dump cannot reproduce below $\approx 10^{-419}$]** *At the nearest conifold $z_3 = 1/\lambda_3$, with $\kappa_0 = 1$ and $\kappa_1 = 0$ ($|\kappa_1| < 10^{-440}$), the computed values are, to 48 decimals with the final digit rounded,*

$$\begin{aligned}
\kappa_2 &= -6.579736267392905745889660666584100756875799604827\dots,\\
\kappa_3 &= 7.599210307330768470917884929095373504836120238934\dots,\\
\kappa_4 &= 13.622344148433291031149701697845733948716693015529\dots,\\
\kappa_5 &= -43.874582810463900205738690933659318347119467057797\dots,\\
\kappa_6 &= 34.702212286332465372786749212708512062179423906148\dots,\\
\kappa_7 &= 51.859399033579894576098101769427996696229062468954\dots,\\
\kappa_8 &= -193.486228134049501553393657261757187130330010383354\dots,\\
\kappa_9 &= 328.247447939140704375272593880584060942588691516576\dots,
\end{aligned}$$

and the first four are identified as

$$\kappa_2 = -4\zeta(2), \qquad \kappa_3 = \tfrac{550}{87}\zeta(3), \qquad \kappa_4 = \tfrac{146}{29}\zeta(2)^2 = \tfrac{365}{29}\zeta(4), \qquad \kappa_5 = \tfrac{514}{87}\zeta(5) - \tfrac{2200}{87}\zeta(2)\zeta(3).$$

The weight-j basis used for the identification is $\{\zeta(2)^a \prod \zeta(\mathrm{odd})\}$, checked for internal relations *before* any relation was read off; see Trap A.1.

5.2. **The primitive normalisation.** Write $\lambda(\varepsilon) = \log \kappa(\varepsilon) = \sum_{j\geq 2} \lambda_j \varepsilon^j$; we call the λ_j the *primitive* Frobenius constants. This is the packaging used in [12, §9].

Corollary 5.2 (A pure $\zeta(5)$). **[PROVED]** *(the relations);* **[VERIFIED: 434 digits]** *(the closed forms).* Assume the identifications of Theorem 5.1. *Then, for the Brown–Zudilin operator,*

$$\lambda_2 = -4\zeta(2), \qquad \lambda_3 = \tfrac{550}{87}\zeta(3), \qquad \lambda_4 = -\tfrac{86}{29}\zeta(2)^2 = -\tfrac{215}{29}\zeta(4), \qquad \boxed{\lambda_5 = \tfrac{514}{87}\zeta(5)}$$

a pure rational multiple of $\zeta(5)$. Moreover $\kappa_5 = \lambda_5 + \lambda_2\lambda_3$ identically, i.e. the entire $\zeta(2)\zeta(3)$ impurity of κ_5 is the decomposable *term $\lambda_2\lambda_3 = (-4\zeta(2))\big(\frac{550}{87}\zeta(3)\big) = -\frac{2200}{87}\zeta(2)\zeta(3)$.*

Proof. $\lambda_2 = \kappa_2$, $\lambda_3 = \kappa_3$, $\lambda_4 = \kappa_4 - \frac{1}{2}\kappa_2^2$ and $\lambda_5 = \kappa_5 - \kappa_2\kappa_3$ from $\log(1 + \sum_{j\geq 2}\kappa_j\varepsilon^j)$. Substituting the values of Theorem 5.1, $\lambda_4 = \frac{146}{29}\zeta(2)^2 - \frac{1}{2}\cdot 16\zeta(2)^2 = -\frac{86}{29}\zeta(2)^2$, and $\lambda_5 = \frac{514}{87}\zeta(5) - \frac{2200}{87}\zeta(2)\zeta(3) + 4\cdot\frac{550}{87}\zeta(2)\zeta(3) = \frac{514}{87}\zeta(5)$. □

Since $m = 5$ for this operator, λ_5 is the primitive part of the *first higher* Frobenius constant. This is the promised occupant of the empty slot (a) of the introduction: no operator in the literature had its first higher Frobenius constant with a pure $\mathbb{Q}\cdot\zeta(5)$ primitive part.

Observation 5.3 (Purity at $j = m$)**. [PROVED]** from the published values, given the identifications of Theorem 5.1 (**[VERIFIED:** 434 **digits]**). For all three operators considered, $\lambda_m \in \mathbb{Q}\cdot\zeta(m)$ at $j = m$:

$$\lambda_2 = -\tfrac{7}{5}\zeta(2)\ (m=2), \qquad \lambda_3 = \tfrac{17}{6}\zeta(3)\ (m=3), \qquad \lambda_5 = \tfrac{514}{87}\zeta(5)\ (m=5).$$

Beyond $j = m$ purity breaks: for the Apéry $\zeta(2)$ operator, where $j = 5$ is three steps past $m = 2$, $\lambda_5 = \zeta(5) - \frac{1}{5}\zeta(2)\zeta(3)$.

5.3. **Beyond $j = 5$: weight inhomogeneity.** From $j = 6$ on, the Brown–Zudilin κ-series stops being weight-homogeneous but stays in the ring of multiple zeta values. This is a positive finding: the Apéry operators, run as controls with the same code, do *not* do it, their κ_j being homogeneous of weight j throughout the published range.

Proposition 5.4. [VERIFIED: 434 digits; tolerance 10^{-380}, coefficient bound 10^{14}]

$$\begin{aligned}
\lambda_6 &= \tfrac{38416}{249777}\zeta(3) + \tfrac{272}{3045}\zeta(2)^3 - \tfrac{392}{7569}\zeta(3)^2,\\
\lambda_7 &= \tfrac{8835680}{8242641}\zeta(3) - \tfrac{38416}{83259}\zeta(2)^2 + \tfrac{784}{2523}\zeta(2)^2\zeta(3) - \tfrac{1070}{87}\zeta(7),\\
\lambda_8 &= \tfrac{1326081904}{272007153}\zeta(3) - \tfrac{8835680}{2747547}\zeta(2)^2 + \tfrac{1319080}{249777}\zeta(5) + \tfrac{74324}{29435}\zeta(2)^4 - \tfrac{26920}{7569}\zeta(3)\zeta(5),\\
\lambda_9 &= \tfrac{4709901251104}{260310845421}\zeta(3) - \tfrac{1326081904}{90669051}\zeta(2)^2 + \tfrac{303388400}{8242641}\zeta(5) - \tfrac{341600}{83259}\zeta(2)^3\\
&\quad - \tfrac{1716176}{1975509}\zeta(3)^2 + \tfrac{48800}{17661}\zeta(2)^3\zeta(3) + \tfrac{385264}{1975509}\zeta(3)^3 + \tfrac{26920}{2523}\zeta(2)^2\zeta(5) + \tfrac{590}{87}\zeta(9).
\end{aligned}$$

The mixing pattern is sharp and reproducible:

Observation 5.5 (Weight pattern)**. [VERIFIED: $j \leq 9$]** λ_j contains exactly the weights j and $3, 4, \ldots, j-3$. The weights $j-1$, $j-2$ and $0, 1, 2$ never occur. The low-weight coefficients propagate along *fixed* weights with constant ratios: writing $c_w(\lambda_j)$ for the weight-w coefficient,

$$c_4(\lambda_j) = -3\,c_3(\lambda_{j-1}) \quad (j = 7, 8, 9), \qquad c_5(\lambda_j) = -\tfrac{3365}{294}\,c_4(\lambda_{j-1}) \quad (j = 8, 9),$$

both exactly. The ratio is *not* constant along the moving top off-diagonal $c_{j-3}(\lambda_j)/c_{j-4}(\lambda_{j-1})$, which is -3 at $j = 7$ but $-3365/294$ at $j = 8$; the two statements coincide only at $j = 7$. All denominators are $\{3, 5, 7, 11, 29\}$-smooth, with 29 and $87 = 3\cdot 29$ ubiquitous.

The mechanism producing this inhomogeneity is **[OPEN]** and is, in our judgement, the sharpest remaining question raised by this computation. It probably requires the polynomial $R(T)$ of **[QUOTED]** [3, Thm. 30] for a non-cyclic σ_0; [3, Cor. 33] assumes a single indicial root, which is not the situation here.

6. What the κ-vector is not

Exclusions are reported with the same care as identifications. Every entry below is a statement about a finite search box.

6.1. **Structural hypotheses. [EXCLUDED: tolerance 10^{-195}, coefficient bound 10^{12}]** The following natural shapes for κ, for the Brown–Zudilin operator:

(a) $\kappa = N \cdot G$ with $N(s) = I(s)/(I_5 s^5)$ the rational indicial factor and G weight-graded — the natural shape suggested by [3, Thm. 30]. The refutation has a clean reason: $\lambda_2, \dots, \lambda_5$ carry no rational part, whereas N has a nonzero linear coefficient.
(b) $\log \kappa$ involving Hurwitz values $\zeta(k, \rho)$ at the other local exponents at $z = 0$ (the hypergeometric shape of [3, Prop. 26]).
(c) powers of $\log c$.

[EXCLUDED: tolerance 10^{-380}, coefficient bound 10^{14}] κ_j and λ_j in the *graded* weight-j basis, for $j = 6, \dots, 16$.

6.2. **The gamma-deformation test.** It is natural to ask whether the Brown–Zudilin operator is a gamma-class deformation of a known Apéry operator, given that their κ_5's all lie in the two-dimensional space $\langle \zeta(5), \zeta(2)\zeta(3) \rangle$. The answer is no, and the reason is instructive.

Observation 6.1. [PROVED] (the linear algebra), given the identification of $\kappa_5(\mathrm{BZ})$ in Theorem 5.1 (**[VERIFIED: 434 digits]**). The identity

$$174\,\kappa_5(\mathrm{BZ}) + 987\,\kappa_5(\text{Apéry } \zeta_3) - 3331\,\kappa_5(\text{Apéry } \zeta_2) = 0$$

holds exactly: in coordinates, $174 \cdot \frac{514}{87} + 987 \cdot \frac{7}{3} - 3331 = 0$ and $174 \cdot \left(-\frac{2200}{87}\right) + 987 \cdot \left(-\frac{17}{3}\right) + 3331 \cdot 3 = 0$. It is also *cheap*: three vectors in a two-dimensional space are always dependent. We record it because such an identity, found numerically and reported without this remark, would look like a discovery.

The content is in the individual identifications, whose $\zeta(2)\zeta(3) : \zeta(5)$ ratios are

$$-\tfrac{1100}{257}\ (\mathrm{BZ}), \qquad -\tfrac{17}{7}\ (\text{Apéry } \zeta_3), \qquad -3\ (\text{Apéry } \zeta_2),$$

against $+2$ for Brown–Zudilin's top period $2[\zeta(5) + 2\zeta(2)\zeta(3)]$.

[EXCLUDED: tolerance 10^{-193}, coefficient bound 10^{14}] A two-term rational relation between $\kappa_j(\mathrm{BZ})$ and $\kappa_j(\text{Apéry } \zeta_3)$, for $j = 5, \dots, 13$. (The hits at $j = 2, 3, 4$ — $\kappa_2(\mathrm{BZ}) = 2\kappa_2$, $493\kappa_3(\mathrm{BZ}) = 1100\kappa_3$, $58\kappa_4(\mathrm{BZ}) = 365\kappa_4$ — are forced, each weight-j space being one-dimensional for $j \le 4$.) The three operators are *not* gamma-deformations of one another in any two-term sense.

Observation 6.2 (The right statement is the primitive one)**. [PROVED]** (the decomposition $\kappa_5 = \lambda_5 + \lambda_2\lambda_3$, valid for any log-series); the BZ row given the identifications of Theorem 5.1 (**[VERIFIED: 434 digits]**), the other two from the published values. In every case $\kappa_5 = \lambda_5 + \lambda_2\lambda_3$, with

$$(\lambda_2, \lambda_3) = \left(-4\zeta_2, \tfrac{550}{87}\zeta_3\right), \quad \left(-2\zeta_2, \tfrac{17}{6}\zeta_3\right), \quad \left(-\tfrac{7}{5}\zeta_2, 2\zeta_3\right)$$

for BZ, Apéry $\zeta(3)$, Apéry $\zeta(2)$ respectively. The space $\langle \zeta(5), \zeta(2)\zeta(3) \rangle$ is a plane of $(\lambda_5, \lambda_2\lambda_3)$ coordinates, not a plane of new periods.

6.3. **Reciprocal periods. [EXCLUDED: tolerance 10^{-193}, coefficient bound 10^{10}]** $\kappa_j(\mathrm{BZ}) \in \mathrm{span}_{\mathbb{Q}}\{\pi^{-2k} : 0 \le k \le 4\}$ for $j = 2, \dots, 8$. The lesson of §7 — always adjoin π^{-2k} to a weight-zero basis — was applied here and returned nothing. This is consistent: the reciprocal period lives in the connection-constant *ratio* c, not in κ, matching [3, Cor. 31], which is a statement about ratios.

7. The purity defect $c = -3/\pi^2$

7.1. **Statement and proof.** The three characteristic rays of (10) carry, respectively, Q_n (growth λ_3^n), the pair I', I'' (decay $|\lambda_2|^n$), and the combination I (decay λ_1^n). The last is the fastest, and it is the one an irrationality proof would want; but it is occupied by an impure period. The constant that measures the exile is

$$c \;=\; \lim_{n\to\infty} \frac{I''_n}{I'_n},$$

equivalently the unique scalar for which $\widehat{I} - cI'$ lands on the minimal ray.

Theorem 7.1. [PROVED]

$$c \;=\; -\frac{1}{2\zeta(2)} \;=\; -\frac{3}{\pi^2} \;=\; -0.3039635509270133143316383896291829167130763240 1674\ldots,$$

and the minimal-ray class it defines is, up to the factor $4\zeta(2)$, the cellular integral itself:

$$M_n := \widehat{I}_n - c\, I'_n = \frac{I_n}{4\zeta(2)}.$$

Proof. Brown and Zudilin prove, for the whole admissible eight-parameter family and not merely at the symmetric point, the decomposition $I(\boldsymbol{a}\cdot n) = 2I'(\boldsymbol{a}\cdot n) + 4\zeta(2)I''(\boldsymbol{a}\cdot n)$ **[QUOTED]** [6, §2 and §7], together with $|\lambda_1(\boldsymbol{a})| < |\lambda_2(\boldsymbol{a})|$, checkable at each admissible $\boldsymbol{a}$ **[QUOTED]** [6, Remark 2]. Let $V_1 = \{u : \limsup_n \log|u_n|/n \le \log|\lambda_1|\}$ be the minimal-ray space; $\dim V_1 = 1$, and $I \in V_1$ while $I', \widehat{I} \notin V_1$. Rewriting the decomposition,

$$I_n = 4\zeta(2)\Big(\widehat{I}_n + \frac{I'_n}{2\zeta(2)}\Big) \in V_1,$$

so $\widehat{I} + I'/(2\zeta(2)) \in V_1$ already. If also $\widehat{I} - cI' \in V_1$, subtracting gives $\big(c + 1/(2\zeta(2))\big)I' \in V_1$; since $I' \notin V_1$ this forces $c = -1/(2\zeta(2))$. The limit form follows because $\widehat{I}_n - cI'_n = O(\lambda_1^n \operatorname{poly} n)$ while $I'_n \asymp \lambda_2^n$ and $|\lambda_1| < |\lambda_2|$. □

Corollary 7.2. [PROVED]$c(\boldsymbol{a}) = -3/\pi^2$ *for* every *admissible cone point $\boldsymbol{a}$. The proof uses only the family-wide decomposition and $|\lambda_1| < |\lambda_2|$. In particular c carries no information about $\boldsymbol{a}$: purity is not deformable within the family.*

7.2. **The certificate.** Independently of the proof, from the exact ladders ($n \le 360$, exact rational arithmetic) at dps $= 2600$, with $c^* := -3/\pi^2$ taken exactly:

Table 3. Convergence of $\widehat{I}_n/I'_n$ to $-3/\pi^2$. **[VERIFIED: 441 digits at $n = 360$]**

n	50	100	150	200	250	300	360
$-\log_{10}\lvert\widehat{I}_n/I'_n - c^*\rvert$	61.367	122.713	184.060	245.408	306.756	368.104	**441.722**

Agreement grows linearly at slope 1.22701 digits per n. The predicted slope is $\log_{10}(\lambda_2/\lambda_1) = \log_{10}(16.86411) = 1.226963\ldots$, and the two agree to 4 digits, so the convergence rate is the predicted $(\lambda_1/\lambda_2)^n$. This is not a numerical coincidence at 441 digits; it is the theorem's own error term. **[PROVED]** (the slope), **[VERIFIED: 441 digits]** (the value). A third-ray check with c^* exact gives $\log|M_n|/n \to \log|\lambda_1| = -5.29756$ from below with the standard n^α correction, and $I_n = 4\zeta(2)M_n$ to full working precision (relative difference $\sim 10^{-2160}$ at $n = 360$).

7.3. **Why a 600-digit search missed it.** Earlier exclusions for c — over a fourteen-term weight-graded basis through weight 8, over $\{1, \log 2, \mathrm{Catalan}, \pi, \pi^2\}$, and others — were all correct. The basis was simply missing the inverse period. The constant c has weight -2, and an integer-relation search over a $\mathbb{Q}$-basis of periods can never see a reciprocal period: $1/\zeta(2)$ lies in the $\mathbb{Q}$-span neither of

$$\{1, \zeta_2, \zeta_3, \zeta_2^2, \zeta_5, \zeta_2\zeta_3\} \qquad \text{nor of the weight-zero ratios} \qquad \{1, \tfrac{\zeta_3}{\zeta_5}, \tfrac{\zeta_2\zeta_3}{\zeta_5}, \tfrac{\zeta_2^2}{\zeta_5}, \tfrac{\zeta_7}{\zeta_2\zeta_5}\}.$$

The one test that would have found it, a two-term relation between $c\,\zeta(2)$ and 1, was never run.

Remark 7.3 (Methodological lesson). Adjoin $1/\zeta(2) = 6/\pi^2$, and more generally π^{-2k}, to any weight-zero integer-relation basis. We state this because it cost a substantial computation, and because it generalises: the constants attached to *ratios* of connection data are exactly the place where [3, Cor. 31]'s "periods with $2\pi i$ inverted" becomes operative.

7.4. **The Tate reading.** Since $4\zeta(2) = -(2\pi i)^2/6$,

$$c \;=\; \frac{12}{(2\pi i)^2}, \tag{12}$$

[PROVED] (exact). This is precisely the shape predicted by **[QUOTED]** [3, Cor. 31] for connection constants of a Picard–Fuchs operator: a period with $2\pi i$ inverted. The defect is a rational multiple of a negative Tate twist — weight -2, depth 0, the most degenerate possible answer. It carries no motivic information beyond the Tate line, and the search for an exotic new period was searching for something that was never there. The graded picture is:

TABLE 4. The three rays of the Brown–Zudilin recurrence. **[PROVED]**

ray	eigenvalue	class	period carried
top (λ_3)	$592.079\ldots^n$	Q_n	$\mathbb{Q}(0)$
middle (λ_2)	$(-0.0843843\ldots)^n$	$I', \widehat{I}$ (span)	$\zeta(5)$ and $\zeta(3)$ separately
minimal (λ_1)	$0.00500378\ldots^n$	$I = 2I' + 4\zeta(2)\widehat{I}$	$2\zeta(5) + 4\zeta(2)\zeta(3)$

The class exiled from the minimal ray is exiled *by* $\zeta(2)$. This is the same object responsible for the failure of a weight-5 harmonic-monomial decomposition of P_n: Brown and Zudilin's top period $\zeta(5) + 2\zeta(2)\zeta(3)$ has depth 2, and its impurity is what blocks the elementary decomposition. The rotation c is the $\zeta(2)$-twist and nothing more.

8. Rate–purity conservation

8.1. **The exact gap.** From the exact roots of $4\lambda^3 - 2368\lambda^2 - 188\lambda + 1$:

$$\text{rate of the minimal ray} = -\log|\lambda_1| = 5.29756135300906258108914228324665 28\ldots$$
$$\text{rate of the middle ray} = -\log|\lambda_2| = 2.47237372274663996054995386721171 15\ldots$$
$$\textit{purity cost} \quad \Delta = \log|\lambda_2/\lambda_1| = 2.82518763026242262053918841603494 13\ldots$$

[VERIFIED: 35 digits]. Note $\Delta = \log 16.86411\ldots$ and $\Delta/\log 10 = 1.226963\ldots$, which is exactly the digits-per-n slope of the 441-digit certificate of Table 3: the same number, now to 34 digits. Δ is the number of nats one must pay to move a linear form from the middle ray to the minimal ray.

TABLE 5. (RPC) on four Apéry-like families. All multiplicities are exact symbolic computations. **[PROVED]** for each row's m and r; the law itself is Conjecture 8.1. The last two columns are columns 2 and 3 re-expressed (Remark 8.2), not independent checks; the substantive content of (RPC-4) is $\lambda_m \in \mathbb{Q}\cdot\zeta(m)$, which is Observation 5.3.

family	m	$1+\lfloor m/2\rfloor$	actual r	depth-≤ 2 monomials of κ_m	$r-1$
Apéry $\zeta(2)$	2	2	2 ✓	$\lambda_2 \to 1$	1 ✓
Apéry $\zeta(3)$	3	2	2 ✓	$\lambda_3 \to 1$	1 ✓
Brown–Zudilin $\zeta(5)$, order 9	**5**	3	**3** ✓	$\lambda_5+\lambda_2\lambda_3 \to 2$	2 ✓
weight-7 cellular, order-4 rec.	**7**	4	**4** ✓	$\lambda_7+\lambda_2\lambda_5+\lambda_3\lambda_4 \to 3$	3 ✓

Row 1: $D^2-t(11D^2+11D+3)-t^2(D+1)^2$. Row 2: $D^3-t(34D^3+51D^2+27D+5)+t^2(D+1)^3$. Row 4: degree 19.

8.2. **The law.** We state the following as a conjecture with explicit falsification conditions, because that is what it is: a pattern read off three families and then successfully predicted on a fourth.

Conjecture 8.1 (Rate–Purity Conservation, (RPC))**.** *Let L be an Apéry-like operator, m the multiplicity of the local exponent* 0 *in the indicial polynomial at $z=0$, and r the order of the associated recurrence.*

(RPC-1) *The characteristic rays are indexed by the* depth $d=0,1,\dots,\lfloor m/2\rfloor$ *of the period they carry; consequently*

$$\boxed{r = 1+\lfloor m/2\rfloor}$$

(RPC-2) *Depth and rate are anti-correlated: $|\lambda_{(d)}|$ is strictly decreasing in d, and the price of purifying from depth d to depth $d-1$ is $\Delta_d = \log\bigl|\lambda_{(d-1)}/\lambda_{(d)}\bigr| > 0$.*

(RPC-3) *The constant effecting the depth-1 to depth-2 raise is the middle-ray connection ratio $c = A_{\widehat{I}}/A_{I'}$, a Tate class $12/(2\pi i)^2$ of weight -2, constant on the whole admissible cone. Hence purity is not deformable while Δ is: no motion in the cone converts rate into purity.*

(RPC-4) (Frobenius witness.) *At the nearest conifold, $\lambda_m \in \mathbb{Q}\cdot\zeta(m)$: the primitive part of the Frobenius constant at $j=m$ is a rational multiple of $\zeta(m)$, so that $\kappa_m = \lambda_m + \sum_{a+b=m}\lambda_a\lambda_b + (depth \ge 3)$ has all of its impurity in the decomposable terms.*

Remark 8.2 (what (RPC-4) does *not* say). It is tempting to add to (RPC-4) that the number of depth-≤ 2 monomials of κ_m equals $r-1$. That would carry no content beyond (RPC-1). For *any* series $\log\kappa = \sum_{j\ge 2}\lambda_j\varepsilon^j$ whatever, the depth-≤ 2 part of $[\varepsilon^m]\exp\bigl(\sum_j \lambda_j\varepsilon^j\bigr)$ consists of the single monomial λ_m together with the pairs $\lambda_a\lambda_b$, $2\le a\le b$, $a+b=m$, of which there are $\lfloor m/2\rfloor - 1$; so the count is $\lfloor m/2\rfloor$ identically, for every operator and with no computation. Asserting that it equals $r-1$ is therefore exactly the assertion $r = 1+\lfloor m/2\rfloor$, i.e. (RPC-1) again. We record the monomials in Table 5 because they are the concrete shape of the witness, not because they are a second test.

Part (RPC-3) is **[PROVED]** for the Brown–Zudilin family (Theorem 7.1 and Corollary 7.2). Parts (RPC-1), (RPC-2), (RPC-4) are the conjectural content.

8.3. **Verification.** The fourth row is a genuine prediction. The law was read off $m=2,3,5$; the multiplicity m for the weight-7 operator was then computed for the first time, from the certified order-4, degree-19 recurrence of the weight-7 family, via $q_j(x) = c_{4-j}(x+j-4)$. Its indicial polynomial factors exactly as

$$\begin{aligned} I(s) = s^7\,(s-1)^3\,(2s-5)\bigl(&3690864s^8 - 59053824s^7 + 409061716s^6 - 1601726448s^5 \\ &+ 3876913912s^4 - 5940014784s^3 + 5627137052s^2 - 3014681200s + 699679047\bigr), \end{aligned} \tag{13}$$

[PROVED] (exact factorisation over $\mathbb{Q}$). Hence $m = 7$ — the multiplicity of the exponent 0 equals the weight, as for $m = 2, 3, 5$ — and $1 + \lfloor 7/2 \rfloor = 4 = r$. ✓

Observation 8.3 (m equals the motivic weight)**. [PROVED]** (on four families.) In every family examined, the multiplicity of the exponent 0 at $z = 0$ equals the motivic weight of the family: $m = 2, 3, 5, 7$ for weights $2, 3, 5, 7$.

8.4. **How to falsify.** A fifth Apéry-like family of weight w falsifies (RPC) if any of the following occurs.

(F1) Its recurrence order satisfies $r \neq 1 + \lfloor w/2 \rfloor$ (falsifies RPC-1).
(F2) The λ_w of its nearest conifold is not in $\mathbb{Q} \cdot \zeta(w)$ (falsifies RPC-4).
(F3) Its middle-ray connection ratio is a non-Tate period (falsifies RPC-3).
(F4) Its characteristic moduli fail to be strictly ordered by depth (falsifies RPC-2).

By Remark 8.2 there is no fifth route through the monomial count: that count is a formal identity and can never fire independently of (F1). Concrete open predictions: a weight-4 family must have $r = 3$; a weight-6 family $r = 4$; a weight-9 family $r = 5$ together with $\lambda_9 \in \mathbb{Q} \cdot \zeta(9)$ — note that the shape $\kappa_9 = \lambda_9 + \lambda_2\lambda_7 + \lambda_3\lambda_6 + \lambda_4\lambda_5 +$ (deeper) is itself formal and predicts nothing. Each of these is a finite computation on any family for which a recurrence is available.

8.5. **Why this is a rate-plus-purity law.** The reason $\zeta(5)$ is out of reach by the Brown–Zudilin family becomes a counting statement. Weight 5 admits $\lfloor 5/2 \rfloor = 2$ depths, so the recurrence must have 3 rays; the pure forms $I', \widehat{I}$ are forced onto the depth-1 ray, and the depth-2 combination $I = 2I' + 4\zeta(2)\widehat{I}$ monopolises the minimal ray. The coefficient $4\zeta(2)$ is $-(2\pi i)^2/6$, and its inverse $c = 12/(2\pi i)^2$ is the cone-constant defect of §7. The rate one would need lives on the ray whose period is impure *by construction of the depth filtration*, and $\Delta = 2.8252\ldots$ is the exact toll.

9. Connection constants

9.1. **High-precision values.** Exact rational ladders $Q_n, P_n, \widehat{P}_n$ were propagated to $n = 520$ from (10) with the anchors of [6]; the linear forms $I' = Q\zeta(5) - P$, $\widehat{I} = Q\zeta(3) - \widehat{P}$, $I = 2I' + 4\zeta(2)\widehat{I}$ were formed at dps $= 3200$ (the cancellation in I is $(\lambda_3/|\lambda_2|)^n = 10^{1999}$ at $n = 520$, which is why so much working precision is needed); the Birkhoff–Stokes limit was then taken on each ray at dps $= 400$, $M = 200$. All four exponents came out $\alpha = -5/2$ automatically, as Proposition 4.2 requires.

Table 6. Connection (Stokes) constants of the Brown–Zudilin family, $u_n \sim A\lambda^n n^{-5/2}$.

constant	value	agreement across n
A_Q	$0.066676425727156767841653340639345444469628745 59\ldots$	322 digits
$A_{I'}$	$-0.75135587647498929950939202170801814994076813034\ldots$	288 digits
$A_{\widehat{I}}$	$0.22838480022321613498452097882537506442071493158\ldots$	288 digits
A_I	$4.78381719294327891215233374724085460054978566494\ldots$	287 digits

As an independent confirmation of Theorem 7.1, $A_{\widehat{I}}/A_{I'} + 3/\pi^2 = 5.7 \times 10^{-381}$: the purity defect reconfirmed to 381 digits by a method sharing no code with the certificate of Table 3. **[VERIFIED: 381 digits]**

At the *first* singularity $z_3 = 1/\lambda_3$ one has $A_P/A_Q = \zeta(5)$ and $A_{\widehat{P}}/A_Q = \zeta(3)$ **[VERIFIED: 3878 digits]**. We flag these as formally trivial — they say $\lim P_n/Q_n = \zeta(5)$, which holds because $I'_n \to 0$ — and record them only as a very strong consistency check on the extrapolation machinery, not as a theorem. Reading the singularities outward from $z = 0$, the connection ratios are

$$z_3 : \ \zeta(5),\ \zeta(3); \qquad z_2 : \ 1/(2\zeta(2)); \qquad z_1 : \ \text{—},$$

the weights descending 5, 3, and then *inverting* to -2. That inversion is the defect.

9.2. **The identifications.** The relation finder was validated first on $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$, on the $\Gamma(1/6)$ duplication formula, and on the Apéry $\zeta(3)$ Stokes constant $S(0) = (1 + \sqrt{2})^2/(2^{9/4}\pi^{3/2})$, before being applied to unknown targets. Basis hygiene is discussed in Trap A.1.

Theorem 9.1 (The Stokes determinant)**. [VERIFIED: 260 digits; relative difference 1.1×10^{-260}]**

$$A_Q \cdot A_{I'} \cdot A_I \ = \ -\frac{\pi^{5/2}}{12\sqrt{37}}, \qquad \textit{equivalently} \qquad \big(A_Q A_{I'} A_I\big)^2 = \frac{\pi^5}{5328}, \quad 5328 = 2^4 \cdot 3^2 \cdot 37.$$

Using $c = -3/\pi^2$ this gives, independently verified, $A_Q \cdot A_{\widehat{I}} \cdot A_I = \sqrt{\pi}/(4\sqrt{37})$.

Theorem 9.2 (The individual constants)**.** *Let $\lambda_3 = 592.0793805346115628\ldots$ be the dominant characteristic root. Then*

$$64\,\big(A_Q\,\pi^{5/2}\big)^2 = u, \qquad 37u^3 - 3219u^2 - 229u - 1 = 0, \qquad u = \frac{-700\lambda_3^2 + 526604\lambda_3 - 6199}{762533}, \tag{14}$$

with $762533 = 37^2 \cdot 557$; that is, $A_Q = \sqrt{u}\,/\,(8\pi^{5/2})$ with $u \in \mathbb{Q}(\lambda_3)$ explicitly. **[VERIFIED: 261 digits]** *Moreover, writing t for the indicated square,*

$$\begin{aligned} t = \big(A_{I'}\pi^{-5/2}\big)^2 : &\quad 1726272\,t^3 + 4171824\,t^2 - 8244\,t + 1 = 0,\\ t = \big(A_I\pi^{-5/2}\big)^2 : &\quad 37\,t^3 + 51504\,t^2 - 58624\,t + 4096 = 0,\\ t = \big(A_{\widehat{I}}\pi^{-1/2}\big)^2 : &\quad 2368\,t^3 + 51504\,t^2 - 916\,t + 1 = 0. \end{aligned}$$

[VERIFIED: all at tolerance 10^{-230}, coefficient bound 10^{16}]

Internal consistency, which is a genuine check and not a restatement:

- The $A_{I'}$ cubic is the $A_{\widehat{I}}$ cubic under $t \mapsto t/9$ — forced by $A_{I'} = A_{\widehat{I}}/c = -(\pi^2/3)A_{\widehat{I}}$. **[PROVED]** (exact polynomial identity).
- The three π-powers $-\frac{5}{2}, +\frac{5}{2}, +\frac{5}{2}$ of $A_Q, A_{I'}, A_I$ sum to $\frac{5}{2}$, matching Theorem 9.1; the other triple $A_Q, A_{\widehat{I}}, A_I$ has powers $-\frac{5}{2}, +\frac{1}{2}, +\frac{5}{2}$ summing to $\frac{1}{2}$, matching $\sqrt{\pi}/(4\sqrt{37})$. **[PROVED]**
- The leading coefficient 2368 of the $A_{\widehat{I}}$ cubic is the middle coefficient of the characteristic polynomial $4\lambda^3 - 2368\lambda^2 - 188\lambda + 1$; the primes 37 and 557 are those of $41218 = 2 \cdot 37 \cdot 557$; the singular cubic has discriminant $2^4 \cdot 37^3 \cdot 557^2$ and $\mathrm{disc}(37u^3 - 3219u^2 - 229u - 1) = 2^4 \cdot 37 \cdot 26357^2$ — the *same square class* 37, i.e. the same quadratic resolvent $\mathbb{Q}(\sqrt{37})$, and indeed $u \in \mathbb{Q}(\lambda_3)$ explicitly. **[PROVED]**

9.3. **The Γ-content.** **[EXCLUDED: tolerance 10^{-240}, coefficient bound 10^6]** The quantity $\log|A_Q|$ lies in the $\mathbb{Q}$-span of

$$\{\log\Gamma(k/s)\} \;\cup\; \{\log\pi,\ \log 2,\ \log 3,\ \log 37,\ \log 557,\ \log|z_1|,\ \log|z_2|\},$$

for $s \in \{3,4,5,6,8,12,24\}$ and combinations thereof.

[EXCLUDED: coefficient bound 10^{20}] $A_Q\pi^e \in \mathbb{Q}(\lambda_3)$ — as opposed to its quadratic extension — for $e \in \{-3/2,-1/2,0,1/2,3/2\}$.

Observation 9.3 (Verdict)**.** **[EXCLUDED: tolerance 10^{-240}, coefficient bound 10^6, $s \in \{3,4,5,6,8,12,24\}$]** The only Γ-value found in the Brown–Zudilin connection constants is $\Gamma(1/2) = \sqrt{\pi}$. The half-integer power is $\pi^{\rho+1}$ with $\rho = 3/2$ the conifold exponent — the classical transfer factor $1/\Gamma(-\rho)$, $\Gamma(-3/2) = 4\sqrt{\pi}/3$ — and the algebraic part is a square root in the S_3 cubic field $\mathbb{Q}(\lambda_3)$ of the singular locus. Within that box: no $\Gamma(r/s)$ with $s > 2$, no cubic-field regulator, no new transcendental. As Remark 9.4 insists, this is a statement about a box; the denominators $s \notin \{3,4,5,6,8,12,24\}$ were not searched.

Remark 9.4 (An earlier negative, corrected). A previous search reported that $A_Q, A_{I'}, A_I$ are "not algebraic times π^k", on the strength of tests of $A\pi^e$ for $e \in \{-\frac{1}{2},0,\frac{1}{2},1,\frac{3}{2},2,\frac{5}{2},3\}$ with minimal polynomials of degree ≤ 12 and coefficient bound 10^5. Those exclusions were correct as stated and simply under-ranged in two ways: the exponents $e = -5/2$ and $e = -1/2$ — the right ones for $A_{I'}, A_I, A_{\widehat{I}}$ — were never tested, and for A_Q the correct exponent $e = +5/2$ *was* tested, but the minimal polynomial (14) has coefficients up to 1.3×10^7, some 130 times the bound used. We record this because it is the standard failure mode of exclusion results and is worth stating explicitly: an exclusion is a statement about a box.

10. Application: no pair-worthiness at weight 7

We now apply the picture to the question it was built for. Zudilin's theorem that one of $\zeta(5),\zeta(7),\zeta(9),\zeta(11)$ is irrational [24, 25] remains the narrowest window containing $\zeta(5)$; a construction producing a small linear form in $1,\zeta(5),\zeta(7)$ alone would narrow it to a pair. The weight-7 analogue of the Brown–Zudilin family is the natural candidate. We show it does not deliver.

10.1. **Setup.** The weight-7 family has a certified order-4, degree-19 recurrence L with palindromic characteristic polynomial

$$7381728\,x^4 - 46800155520\,x^3 + 501765579072\,x^2 - 46800155520\,x + 7381728,$$

whose roots come in reciprocal pairs **[PROVED]**:

$$\{6329.2605150\ldots,\ 10.6453896153\ldots,\ 0.0939373791041\ldots,\ 1.5799634059 \times 10^{-4}\},$$

with log-rates ± 8.75293868604055 and ± 2.36512689845195. The ladder consists of q_n, s_n, $\widehat{P}_n$ and (unavailable beyond small n) P_n; the linear forms are $I''_n = -9q_n\zeta(5) + 2s_n\zeta(3) - \widehat{P}_n$ and the primitive form I'_n. Write

$$\text{sector A}:\ \rho_A = 1.5799634 \times 10^{-4}, \qquad \text{sector B}:\ \rho_B = 0.0939373791041.$$

Sector A lies below $e^{-7} = 9.1188 \times 10^{-4}$ and would pass the irrationality criterion; sector B does not.

10.2. **The sector theorem.** The earlier argument for this family required two facts: that I'' rides sector B, and that I rides sector A, so that the sector-B coefficient of $I' = I - \zeta_2 I''$ is nonzero. The second is a delicate assertion that a cancellation is exact. It is not needed.

Theorem 10.1 (Sector exclusion by positivity). *Given that $s, \widehat{P} \in \mathrm{Sol}(L)$ (see §10.7), the primitive weight-7 form satisfies*

$$I'_n \;=\; I_n - \zeta_2 I''_n \;=\; I_n + \zeta_2 |I''_n|, \qquad \textit{hence} \qquad |I'_n| \;\geq\; \zeta_2\, |I''_n|,$$

so that $\limsup_n |I'_n|^{1/n} \;\geq\; \rho_B \;=\; 0.0939374\ldots \;\gg\; \rho_A$. *Sector A is* excluded*, not merely disfavoured.*

Proof. I' is a sum of two positive quantities. The two inputs are: $I''_n < 0$ for all $n = 0, \ldots, 140$, verified in exact rational arithmetic **[VERIFIED: exact,** 141 **indices]**; and $I_n > 0$ for all n, which is rigorous, I being an all-positive fourfold series. Given $s, \widehat{P} \in \mathrm{Sol}(L)$, $|I''_n|^{1/n}$ converges to one of the four discrete values above, and §10.3 identifies the limit as ρ_B. No conspiracy of coefficients can make I' decay faster than I''. □

The improvement over the cancellation argument is that the verdict is now independent of everything about I_n's own decay rate, of the denominator of P_3, and of any grid hypothesis. The sole remaining dependency is $s, \widehat{P} \in \mathrm{Sol}(L)$, discussed in §10.7.

10.3. **Measuring the sector.** I''_n was evaluated from exact rationals at dps $= 4.9n + 100$, so there is no forward-propagation instability. The statistic is $-\log |I''_{n+1}/I''_n|$; the model $I''_n \sim C\rho^n n^\beta$ gives the raw estimator a $-\beta/n$ bias, killed by Richardson extrapolation.

TABLE 7. Rate of I''. Target $-\log \rho_B = 2.36512689845$. **[VERIFIED: 5 digits]**

n	raw	Richardson-1 $(n, 2n)$	Richardson-2
10	2.6796806457	2.3818974599	2.3655899615
20	2.5307890528	2.3696668361	2.3651904833
30	2.4775676766	2.3672005606	**2.3651463543**
40	2.4502279445	2.3663095715	—
50	2.4335830363	2.3658901534	—
60	2.4223841186	2.3656599059	—
69	2.4150358618	**2.3655314035**	—

This is a *discrete identification, not a continuous fit*: given $s, \widehat{P} \in \mathrm{Sol}(L)$, the only admissible answers are $\{-8.752939, -2.365127, +2.365127, +8.752939\}$. The Richardson-2 value is within 2×10^{-5} of one of them and 4.7 away from the nearest alternative. Two disjoint windows agree. **Sector B.**

10.4. **The honest denominator exponent.** Whether the true denominator budget is below 7 is the one remaining lever, and it is not available. Measured directly on the 361 exact weight-5 ladder terms — the only family where the true P is known:

$$\mathrm{den}(P_n) \mid 12\, d_n^5 \;\text{ for } 361/361; \qquad \mathrm{den}(P_n) \mid d_n^5 \;\text{ for } 1/361; \qquad \mathrm{den}(P_n) \mid 6\, d_n^5 \;\text{ for } 1/361,$$

[VERIFIED: exact, $n \leq 360$**]**, so the constant 12 is sharp and the exponent is exactly the weight. The approach to the exponent is

$$\frac{\log \mathrm{den}(P_n)/n}{\log(d_n^5)/n} = 0.9953,\; 0.9928,\; 0.9947,\; 0.9936 \quad \text{at } n = 80, 200, 320, 355,$$

converging to 1. The weight-7 descent ladder behaves identically: $\log \mathrm{den}(\widehat{P}_n)/n$ divided by $5 \log d_n/n$ is 0.9856 at $n = 140$, i.e. the descent exponent 5 is tight.

Observation 10.2. **[VERIFIED: as tabulated]** $\kappa = 7$ is the honest denominator budget for the weight-7 primitive form, not a placeholder. There is no $d_n^2 d_{2n}$-type structure lowering it: the analogue that can be measured is tight at its nominal value. (The slack members are $\mathrm{den}(s_n) \sim d_n^{1.95}$ and weight-5 $\mathrm{den}(\widehat{P}_n) \sim d_n^{3.95}$, but the binding constraint on a linear form is its constant term, and that is tight.)

To pass at sector B one would need $\kappa < 2.365$, i.e. $\mathrm{den}(P_n) \lesssim d_n^{2.365}$. The shortfall is $d_n^{4.635} \approx 103^{\,n}$.

10.5. **The worthiness numbers.** With $C_1 = \log \lambda_{\max}$, $C_0 = \log \rho(I')$, $C_2 = \kappa$, the two standard measures are

$$\gamma^{\mathrm{ratio}} = \frac{-C_0}{C_2}, \qquad \gamma^{\mathrm{worth}} = \frac{C_1 - C_0}{C_1 + C_2},$$

which exceed 1 simultaneously, both conditions being equivalent to $-C_0 > C_2$. **[PROVED]**. At weight 5 the second reproduces Brown–Zudilin's printed symmetric-point worthiness $\gamma = 0.777959763\ldots$ **[QUOTED]** [6, §2] **[VERIFIED: 8 digits]**. At weight 7:

TABLE 8. Weight-7 worthiness. $C_1 = 8.75293868604$, $C_2 = \kappa = 7$. **[VERIFIED: 12 digits]**

	$-C_0$	γ^{ratio}	γ^{worth}
sector B (actual)	2.36512689845	0.337875271	0.705777240
sector A (hypothetical)	8.75293868604	1.250419812	1.111276932

The exponent gap is $\kappa - (-C_0) = 4.634873$ nats, a factor $e^{4.634873} = 103.01$ per step. The margin is not close.

10.6. **Rhin–Viola leverage, quantified.** The dimensionless invariant that a Rhin–Viola group [18, 19] moves is $(-C_0)/\kappa$. Measured at weight 5 over the entire known orbit:

$$\text{symmetric point } \boldsymbol{a} = (1,\ldots,1): \quad \frac{-C_0}{\kappa} = \frac{2.47237372}{5} = 0.494475,$$
$$\text{record point } \boldsymbol{a} = (8,16,10,15,12,16,18,13): \quad \frac{-C_0}{\kappa} = \frac{31.55296935}{49.60574813} = 0.636075,$$

[VERIFIED: 6 digits], a maximal leverage factor of 1.286364 (+28.6%). Applying the same relative leverage to weight 7 from sector B gives

$$\gamma_{5,7} \;\approx\; 0.337875 \times 1.286364 \;=\; 0.434631,$$

still less than half of 1. Reaching $\gamma_{5,7} = 1$ requires a factor 2.9597 (+196%). In additive terms, the group would buy 0.6773 nats where 4.6349 are needed — a factor 6.84 on the γ^{ratio} normalisation, or 3.68 on γ^{worth}; either way several times the total leverage the Rhin–Viola group is empirically worth at weight 5. **[VERIFIED: 12 digits]**

Corollary 10.3. *Asymmetric-orbit optimisation cannot close a sector-B gap. It is worth running only if a different sector is in play.*

10.7. **Residual dependency, stated plainly.** Everything in this section is unconditional except the hypothesis $s, \widehat{P} \in \mathrm{Sol}(L)$, which rests on two things.

(i) L is the true operator for q_n: empirically certified on 74 exact terms (70/70 relations exactly zero) but not proved — no creative-telescoping certificate exists. **[VERIFIED: exact, 70 relations] [OPEN]** as a proof.

(ii) An arithmetic certificate. Propagating $s, \widehat{P}, P, q$ exactly in $\mathbb{Q}$ gives the denominator ledger

$$\mathrm{den}(s_n) \mid d_n^5 \ \text{ for } 141/141, \qquad \mathrm{den}(\widehat{P}_n) \mid d_n^5 \ \text{ for } 141/141, \qquad n = 0, \dots, 140.$$

The control experiment propagates three generic rational initial conditions through the same L:

TABLE 9. Denominator ledger versus generic solutions ($\log_{10}$ of denominators). **[VERIFIED: exact]**

n	$\mathrm{den}(s_n)$	$\mathrm{den}(\widehat{P}_n)$	generic-1	generic-2	generic-3	d_n^5
20	14.5	37.6	63.3	64.5	64.1	41.8
40	29.0	74.2	116.3	117.6	117.1	78.6
60	47.2	122.3	180.1	181.4	180.9	124.9

The divisibility den $\mid d_n^5$ holds 61/61 for the true ladder and 4/61 for every generic solution, the four being trivial small-n cases. A degree-19 operator divides by $c_4(n)$ at every step, so a generic solution drags in the prime factors of $c_4(n)$ for all n and its denominator overshoots d_n^5 by some 55 digits at $n = 60$. That the true $(s, \widehat{P})$ remain d_n^5-smooth for 141 consecutive n is evidence for $s, \widehat{P} \in \mathrm{Sol}(L)$ of a strength we regard as decisive given (i), though it is evidence and not proof.

The same test separates P: the L-propagated P satisfies $\mathrm{den}(P_n) \nmid d_n^7$ for 101 of the 138 indices $n = 3, \dots, 140$, the excess taking only the values $\{1, 107, 321 = 3 \cdot 107\}$, with $\mathrm{den}(P_n) \mid 321\, d_n^7$ for 141/141. So the L-propagated P is arithmetically clean but wrong: it obeys a bounded-constant d_n^7 law, yet the persistent prime 107 — absent from $q, s, \widehat{P}$, which are strictly d_n^5-clean — marks it as a different solution branch. The exclusion of P from $\mathrm{Sol}(L)$ does not rest on the 107: the propagated value $I_3' = 43.71$ contradicts $I_3' = I_3 + \zeta_2 |I_3''| = 3.5675 \times 10^{-5}$ to four significant figures, the latter following from a rigorous majorant $0 < I_3 \leq 3.0903 \times 10^{-9}$. **[PROVED]** (given the ladder).

Remark 10.4. If L itself were wrong, the entire characteristic-polynomial framework would collapse, sector A included; so no route to a sector-A pass runs through doubting L. We found no evidence whatsoever pointing toward sector A, and one new structural argument (Theorem 10.1) ruling it out given the ladder.

11. Discussion: what the middle-root obstruction means

We state what we take the results to establish, and, separately, what they do not.

11.1. What is established.

(1) *The Brown–Zudilin operator has a well-defined and computable κ-vector, and its first higher Frobenius constant has a pure $\zeta(5)$ primitive part.* The dictionary between irrationality constructions and the gamma-conjecture circle is therefore usable in both directions at weight 5, and not only at weights 2 and 3.

(2) *The purity defect of the family is Tate.* There is no unnamed constant in the middle of this construction. Theorem 7.1 identifies the obstruction's magnitude ($c = 12/(2\pi i)^2$) and Corollary 7.2 shows it is constant on the entire admissible cone. This closes a line of search rather than opening one, and we regard closing it as the more useful outcome: the eight-parameter family cannot be tuned to remove the impurity, because the impurity does not depend on the parameters.

(3) *The obstruction is a counting phenomenon, conjecturally.* If (RPC) holds, then the number of rays is $1 + \lfloor m/2 \rfloor$ and the assignment of periods to rays is by depth. In that case a cellular family of weight $w \geq 4$ *must* place its purest forms off the minimal ray, and the toll Δ is not a defect of the construction but a structural feature of the depth filtration. The four-family verification (Table 5) is evidence; the falsification conditions (F1)–(F4) are the way to test it, and each is a finite computation.

(4) *At weight* 7 *the same thing happens, and the margin is large.* Theorem 10.1 excludes the fast sector by a positivity argument that needs no cancellation hypothesis. The measured gap is 4.635 nats against a Rhin–Viola leverage empirically worth 0.677 nats at the same normalisation. The weight-7 sector-B verdict is the *generic* outcome of this construction, not bad luck: weight 5 shows the identical dichotomy, and there too the smallest saddle would have proved irrationality while the primitive form provably does not ride it.

11.2. **What is not established.** The results say nothing about whether $\zeta(5)$ or $\zeta(7)$ is irrational, and nothing about whether *some other* construction rides the minimal ray with a pure form. (RPC) as stated constrains cellular and Apéry-like families with a single recurrence and a depth-graded period assignment; it does not preclude a construction outside that class. Nor do we claim the middle-root phenomenon is unavoidable in principle: what we claim is that it is stable across the two weights where it can currently be measured, and that within the eight-parameter cone it cannot be moved.

11.3. **Where the leverage would have to come from.** The measured numbers say where it would not. Orbit optimisation on $\mathcal{M}_{0,10}$ operates on $(-C_0)/\kappa$, and at weight 5 the whole group is worth a factor 1.286 on that invariant; weight 7 needs 2.960. The productive redirection is therefore not orbit search but a construction whose *primitive* form rides the small saddle — that is, an attack on the phenomenon $C_0 = \log|\lambda_{\mathrm{middle}}|$ rather than $\log|\lambda_{\min}|$ itself. Three concrete handles emerge from the computation.

(i) *The inhomogeneity of* $\lambda_j(\mathrm{BZ})$ *for* $j \geq 6$ (Observation 5.5). Its mechanism is **[OPEN]**, and every natural structural hypothesis we could formulate is excluded (§6). Whatever explains the weight pattern $\{j\} \cup \{3, \ldots, j-3\}$ and the fixed-weight propagation constants -3 and $-3365/294$ is a genuine feature of a non-MUM Frobenius point and is not present in the rank-2 literature.

(ii) *The second condition of* [3, Def. 21] for the Brown–Zudilin operator — analyticity at c of the σ_c-invariant solutions of $L^\vee$ — is **[OPEN]**, and its verification would put §5 on a fully rigorous footing rather than the empirical licence of Observation 4.3.

(iii) *A primitive operator* $\widetilde{L}$ annihilating I' at weight 7 remains unhuntable by data fitting: there is no P or I' data to fit beyond $n = 3$, and the residue and high-precision routes are both walled. The one $\widetilde{L}$-adjacent fact now available is that the L-propagated P satisfies $\mathrm{den}(P_n)/d_n^7 \in \{107, 321\}$ constantly, so any $\widetilde{L}$ must differ from L in a way that changes the constant-term arithmetic without changing $q, s, \widehat{P}$.

11.4. **A remark on evidence.** Three of the findings above began as reported negatives that turned out to be under-ranged or mis-based: the "not algebraic times π^k" verdict on the connection constants (Remark 9.4), the exclusion of c from every period basis tried (§7.3), and a spurious integer relation produced by a degenerate basis (Trap A.1). In each case the earlier statement was true of the box that was searched. We have therefore attached to every exclusion in this paper the tolerance and the coefficient bound, and to every identification the digit count, and we recommend the practice.

Appendix A. Two traps

Both cost real time, and both generalise.

Trap A.1 (Degenerate bases in integer-relation search). An integer-relation algorithm returns a relation among the vectors it is given. If the basis has an internal relation, the algorithm will find *that*, with coefficient zero on the target, and the output looks like a discovery.

Two instances occurred here.

(a) *Zeta-value bases.* A weight-≤ 8 basis containing both $\zeta(8)$ and $\zeta(2)^4$ returned the relation $175\zeta(8) = 24\zeta(2)^4$ with coefficient 0 on the target. **[PROVED]** (exact: $\zeta(8) = \frac{24}{175}\zeta(2)^4$.)

(b) *Logarithm bases.* For the Brown–Zudilin singular cubic $z^3 - 188z^2 - 2368z + 4$, the product of the roots is -4, so

$$\log|z_1| + \log|z_2| + \log|z_3| = \log 4$$

[PROVED] (exact). Including all three logarithms in a search basis alongside $\log 2$ is therefore degenerate, and it produced a spurious zero-on-target hit until $\log|z_3|$ was dropped.

Rule. Run the relation finder on the basis alone, first. If it returns anything, the basis is degenerate and any subsequent hit must be re-read.

Trap A.2 (Resonant Frobenius normalisation). The weight-7 operator has indicial polynomial (13), which contains the factor $s^7(s-1)^3$: the exponents 0 and 1 are both present and *resonant*. Consequently $a_n(s)$ has a triple pole at $s = 0$, and the Frobenius deformation is not defined by the naive normalisation $a_0 = 1$.

Clearing the pole with $a_0(s) = s^3$ produces clean-looking values $\lambda_2 = -\frac{292}{55}\zeta(2)$ and $\lambda_3 = \frac{432}{55}\zeta(3)$, and then contaminated values from λ_4 onward: the base point has slid to the exponent-1 solution. *The weight-*7 *κ-values are therefore quarantined: the two displayed in the previous sentence are quoted here as the illustration of the trap and nowhere else, and no result of this paper uses any of them.* The (RPC) test on the weight-7 family (Table 5, row 4) rests only on the exact symbolic facts $m = 7$ and $r = 4$, both of which are independent of the normalisation. Handling the resonance correctly requires the resonant Frobenius basis and is **[OPEN]**.

Appendix B. Computational record

All computations were performed in exact rational arithmetic where the statement is exact (Python `fractions`, `sympy`) and in arbitrary-precision floating point otherwise (`mpmath` with the `gmpy2` backend). The following table summarises the run parameters behind each numerical claim.

claim	parameters
Instrument validation, §3	dps $= 220$, $K = 13$, $M = 130$, $n \in \{300, 400\}$; self-agreement 208–210 digits; residuals $< 10^{-219}$
Errata, Observations 3.2–3.3	as above; discrepancies 4.6×10^{-219} and 5.3×10^{-220} against the corrected forms
κ-vector, Theorem 5.1	dps $= 520$, $K = 16$, $M = 260$, $n \in \{700, 800\}$; self-agreement 434 digits; relation tolerance 10^{-380}, coefficient bound 10^{14}; residuals 10^{-440}–10^{-451} in-session. The archived `kappas_hi.json` is written at 420 digits, so a recheck from the archive alone floors at $\approx 10^{-419}$
Structural exclusions, §6	tolerance 10^{-195}, bound 10^{12} (shapes); tolerance 10^{-380}, bound 10^{14} (graded bases, $j \le 16$)
Gamma-deformation exclusion, §6.2	tolerance 10^{-193}, bound 10^{14}, $j = 5, \dots, 13$
Reciprocal-period exclusion	tolerance 10^{-193}, bound 10^{10}, $j = 2, \dots, 8$
Defect certificate, Table 3	exact ladders $n \le 360$, dps $= 2600$; 441 digits at $n = 360$; slope 1.22701 digits per n
Connection constants, Table 6	exact ladders $n \le 520$; I formed at dps $= 3200$; Stokes limit at dps $= 400$, $M = 200$; 287–322 digits
Connection identifications, §9	Theorem 9.1 at relative difference 1.1×10^{-260}; (14) at 261 digits; cubics at tolerance 10^{-230}, bound 10^{16}
Γ exclusions, Observation 9.3	$s \in \{3, 4, 5, 6, 8, 12, 24\}$, tolerance 10^{-240}, bound 10^6; and bound 10^{20} for $A_Q \pi^e \in \mathbb{Q}(\lambda_3)$
Weight-7 ladder, §10	exact $\mathbb{Q}$ propagation $n = 0, \dots, 140$; I'' evaluated at dps $= 4.9n + 100$; weight-5 denominators on 361 exact terms

Appendix C. Citation discipline

Every reference below was checked against a copy of the source: the bibliographic data were read from the published text or preprint, and where a value or a statement is attributed to a specific location in a source, that location was re-read. Three references were added late, and the following remarks record how they are used:

- [27] — cited only in §2.4, for orientation; nothing in this paper depends on it. The statement attributed to it there is its Theorem 1.4 together with its equation (2), $\log \Gamma_p(x) = \Gamma_p'(0)x - \sum_{m \ge 2} \zeta_p(m) x^m / m$.
- [28] — for the classical theory of formal solutions at an irregular singular point of a difference equation. The construction we use is derived from scratch in Proposition 2.1, so the citation is attributional only.
- [29] — for the integer-relation algorithm. All identifications reported here were re-verified against their closed forms after detection, at the stated precisions, so no claim depends on the algorithm's provenance.

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